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DATA SCIENCE JUNE BATCH

MUSIC RECOMMENDATION AND SONG POPULARITY ANALYSIS USING MACHINE LEARNING

1. Problem Statement

Music streaming platforms contain a large number of songs, making it difficult for listeners to discover tracks that match their musical preferences. This project aims to develop a content-based music recommendation system that suggests similar songs using audio features and a machine learning regression model that estimates song popularity from available song characteristics. The application will also provide exploratory insights into the relationships between audio features and popularity.

2. Machine Learning Type and Target Variable

- **Machine Learning Task:** Regression for popularity prediction, supported by a content-based recommendation system.
- **Target Variable:** `popularity`, the numerical popularity score provided in the dataset.
- **Models:** Dummy Regressor (baseline), Linear Regression, Decision Tree Regressor, and Random Forest Regressor. The recommendation component will use audio-feature similarity.

3. Dataset Source and Size

- **Dataset Name:** Spotify Tracks Dataset.
- **Source:** Kaggle.
- **Dataset Link:** `https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset`
- **Dataset Size:** To be confirmed from the downloaded CSV file after checking the number of rows and columns.

- **Features:** Track name, artist, track genre, popularity, danceability, energy, valence, acousticness, instrumentalness, speechiness, tempo, and loudness, depending on the dataset version.

4. End Users

The intended users are music listeners who want to discover similar songs, as well as music analysts and content-curation teams who want to explore song characteristics and popularity patterns.

5. Success Measurement

The regression models will be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) on held-out test data. Cross-validation will be used for model comparison. Lower MAE and RMSE values and a higher R^2 score will indicate better predictive performance.

6. Technical Architecture

